

The Economy

Seminar 1: Introduction and GDP

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Sciences Po Paris - Reims

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Welcome!

- Who am I?
 - Teaching Fellow / Head TA in Economics, Reims
 - Research Fellow in Behavioral Economics and Applied Microeconometrics, Paris
- **Wednesdays.** 14h30-16h30 / 16h40-18h40 / 18h50-20h50
- No class next week! PS2 due Feb 14th.
- Contact: emirtarik.dakin@sciencespo.fr or office hours (TBD).

Housekeeping

- **Course material:** lecture slides and problem sets on the lecture Moodle; seminar slides and supplementary materials (+ problem sets) on the seminar Moodle.
- This class is **difficult**.
 - Everything covered in lectures, seminars or the reading **COULD** come up in the exam.
 - Our expectations are set very high and it is my job to prepare you for those expectations.
 - Review the lecture **BEFORE** the seminar, do the problem set as best as you can, do the readings.
 - It's okay to not always be able to do everything - but it's important that you try.
 - Ask questions, clarify doubts as soon as you have them. Don't stress about the participation.
- Respect each other's learning trajectories - be patient, be open-minded, be ready to participate but also to listen.

Assessment and grading

There are **four** assessments for your total grade.

- ① 8 graded problem set questions (2/9).
 - 1 pre-assigned question from each of the 10 problem sets excluding PS1 (i.e. PS2-PS11), to be handed in at the start of class.
 - 8 best grades out of 10 chosen, provided you hand in all 10.
 - Each non-submission includes a zero grade in the 8 best.
- ② 5 graded quizzes and participation (2/9).
 - 6 quizzes that I will announce from a week earlier.
 - 5 best grades are chosen.
 - You will get a score between 1-10 for your participation in class.
- ③ Midterm (2/9). March 23rd.
- ④ Final (3/9). TBD.

Exam formats will be released later this semester but expect combination of MCQs, short questions ("course questions"), theory (maths + graphs) and empirical (read graphs + comment).

Assessment and grading

Assessment type	no. of assessments	Grade per assessment	Total grade
PSets	10 questions, 8 best	2.8%	22.2%
Quizzes	6 quizzes, 5 best	3%	15%
Participation	continuous	7.2%	7.2%
Midterm	3h	22.2%	22.2%
Final	3h	33.3%	33.3%

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What is economics for you?

Thinking about economics

Economics as a scope of study: not just about economies (though it is a huge part of it) but about economic phenomena at multiple scales.

- **Country A vs Country B:** do they trade? are they culturally similar? why is there global poverty? why are there failed states?
2024: why is DE's growth rate lower than FR?
- **Within a country:** what causes unemployment to rise? are taxes a good idea? should we install a basic income? who should pay for social security? **2024: should access to social security for foreigners be restricted to those who work?**
- **Within industries:** are we a waning industry? how do we regulate new-comers? how do we promote research? who decides the sectoral wage? why does an engineer earn more than a teacher? **2024: will AI replace workers?**

Thinking about economics

- **Within companies:** how much do we produce? what is our advertising strategy? do we expand? when do we hire and fire? **2024: should we condition gov. subsidies on ecological performance?**
- **In between workers:** do I spend my wage on cigarettes or should I save it? when do I believe my boss if he isn't always telling the truth? what should I buy for groceries? when should I buy a house? why does Sally earn less than James?
- **Within families:** when should we get married? should we have a third child? how do we distribute household chores?

Every single one of these questions has a subfield in economics. Economics is not just about the economy. It's about every interaction between every possible agent in the world we live in. It's just how we approach these interactions that differs.

Thinking about economics

The way we answer those previous questions can be very different but they are often underpinned by a single economic methodology.

Economics as a **methodology**: not unlike the scientific method.

- We develop **theoretical models** of phenomena: relationships between variables.
- Check that the models are corroborated by **real world evidence** (take the data to the model).
- Use the corroborated models to make **predictions** (take the model to the data).

Thinking about economics

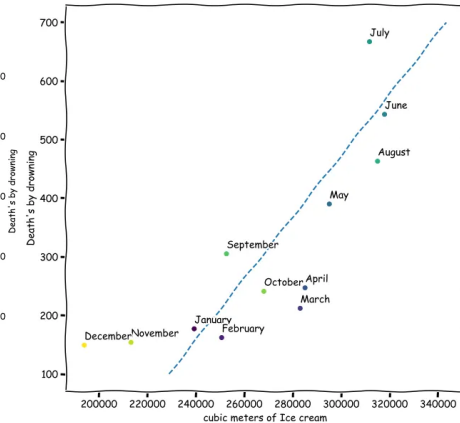
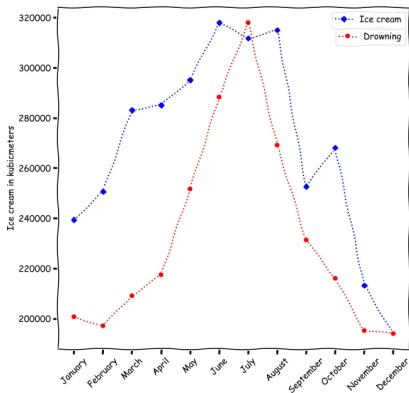
We use "controlled" empirical environments:

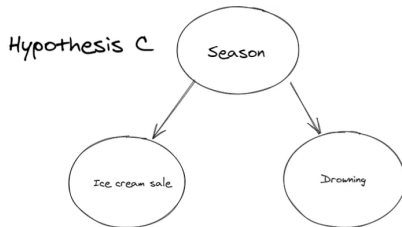
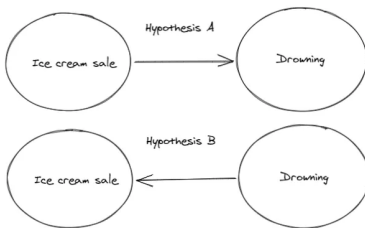
- Experiments in **labs**: behavioural economics → external validity issues?
- Experiments in the **field**: randomised controlled trials (RCTs) → ethical issues?
- **Natural** experiments: minimum wage example - more on this in the PSet.

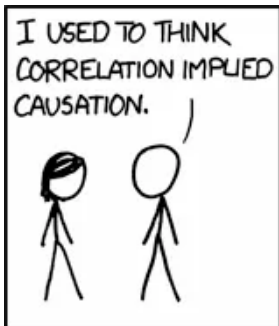
But there's a tradeoff: **control** vs **external validity**.

Even if we do empirical work, we almost always get a bunch of correlations, but a lot more extra justification work (called identification strategies) are needed to identify causalities.

Correlation and causality







PS1 Reading: natural experiments and causality

Group work. Read the text that presents the work of the three winners of the 2021 Sveriges Riksbank Prize in Economics. Choose an example of a natural experiment described in the article and explain:

- What is the question being studied and why does it require the use of a natural experiment?
- Who are the units in the treatment group and the units in the control group?
- What is the result of the experiment?
- How can these results be useful for understanding the functioning of the economy and/or reflecting on the effectiveness of certain policies?

Experiment	Education on Income	Minimum Wage on Unemployment	Immigration on Employment	School Resource on Student Future Success
Study Question + Why Natural Exp?	<i>How much additional year of educ lead to higher earnings → might be that long-studiers are inherently more likely to be high-earners vs. short-studiers + selection into longer studies</i>	<i>Does increase in min. wage lead to lower employment? → possible reverse causality: as unemployment rises, employers set lower wages so calls for min wage increase</i>	<i>Does immigration lead to wage decrease + increased unemployment for natives? → immigrants might select into favourable markets</i>	<i>Does more school resources lead to higher level of future student success? → confounding variable: low school performance might lead to more investment</i>
Treatment and Control Groups	Treatment = extra year of education for those who are born later in calendar year; control group = those born later in calendar year	Treatment group = fast-food workers in NJ that benefited from increase in min wage; control group = fast-food workers in east PA with no change in min wage	Treatment group = native workers in Miami; control group = native workers in other comparable US cities (that had no immigration wave)	Treatment group = people living in city A but grew up in state B with high resources; control group = people living in city A but grew up in state B with low resources
Results	additional year of education → 9% increase in income	increase of min wage by 80 cents → no significant effect on employment	wages did not fall + unemployment did not increase among natives after immigration	higher teacher density in home state leads to higher returns on education
Application to Economy/Policy	e.g. informs schooling policy (compulsory education age)	beyond labour supply effects → possible changes via labour demand? labour market structure?	how to facilitate immigration implantation in nascent industries?	makes education investment less dependent on performance

Three approaches to GDP

GDP is a measure of aggregate observed production of new goods and services over a yearly/quarterly period of time (= a flow variable).

- **Demand / Expenditure approach.**

$$\underbrace{Y}_{\text{GDP}} = \underbrace{C}_{\text{consumption of final goods}} + \underbrace{I}_{\text{investment (capital \& inventory)}} + \underbrace{G}_{\text{govt. spending on final goods}} + \underbrace{NX}_{\text{net export of final goods}} = X \text{ (exports)} - M \text{ (imports)}$$

- **Income approach.**

$$Y = \underbrace{\text{capital income}}_{\text{firm profit or capital rent}} + \underbrace{\text{labour income}}_{\text{wages paid to workers}}$$

- **Value-added approach.**

$$Y = \sum_{\text{for each producer}} \text{all the added value in that producer's final good relative to intermediate inputs}$$

where VA is sale price of final good - the costs of intermediate inputs (for **each producer**).

GDP and GNP: Domestic, National and Foreign Productions

GDP focuses on **domestic production** and **domestic consumption**, regardless of the nationality of the producers and consumers:

- for the expenditure approach: we add **domestic consumption of domestic goods** in C , **foreign consumption of domestic goods** in X and we subtract **domestic consumption of foreign goods** in M .
- for the income approach: we only count the capital income and labour income paid earned domestically.
- for the value-added approach: we consider the added value in each domestic producer's production, subtracting intermediate inputs from both domestic and foreign sources.
→ so Peugeot's VA would be the sale value of the car - the costs of all the component parts from France and from abroad.

GDP and GNP: Domestic, National and Foreign Productions

GNP (gross national product) focuses on **production and consumption by a country's nationals/citizens**, regardless of where they are located.

- relative to GDP, this excludes any income and domestic transfers paid out to foreigners but includes foreign income and foreign transfers paid out to natives abroad.
- transfers include profits, remittances, interest payments, dividends, grants.

GNP/GDP ratio

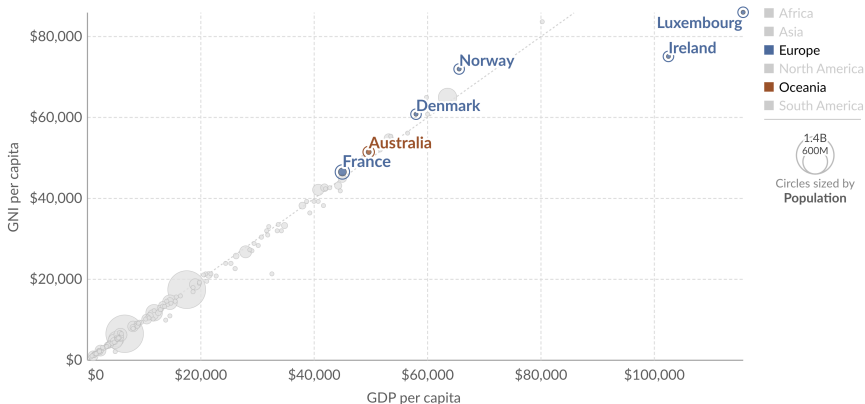
$\text{GNP/GDP} > 1$ implies: $\text{GNP} > \text{GDP}$

- $\text{GNP} < \text{GDP}$ ratio below 1: implies the net income from abroad is negative → **Ireland**. Why?
- $\text{GNP} > \text{GDP}$ ratio above 1: implies the net income from abroad is positive → **Norway**. Why?

GNI per capita vs. GDP per capita, 2021

Our World
in Data

Gross national income (GNI) measures the total income earned by residents of a country, including income earned abroad. Gross Domestic Product (GDP) instead counts only income generated from goods and services produced domestically.



Data source: Data compiled from multiple sources by World Bank; World Bank (2023)

Note: This data is expressed in international-\$¹ at 2017 prices to account for inflation and differences in the cost of living between countries.

OurWorldInData.org/economic-growth | CC BY

Comparing GDP across countries

Huge variation in both **GDP levels** and **GDP growth rates** across countries: different demographics, different economic structures, different measurement methods, different prices.

- ① **GDP per capita**. To weed out **demographic differences**.

$$\frac{Y}{N} \quad \text{where } N \text{ is population}$$

- ② **real GDP**. To weed out **price changes** over time.

$$rGDP = \frac{GDP}{P_{now}/P_{ref.yr}}$$

- ③ **PPP (purchasing power parity) adjusted GDP**. To weed out **disparities in spending power**.

$$GDP_{ppp} = GDP \times \frac{\text{price in ref. country}}{\text{price in country in question}}$$

PS1 Exercise 1. Three measures of GDP

Consider a fictitious economy producing cotton-based T-shirts. There is a cotton farmer and a T-shirt producer. Producing one T-shirt requires 1.5kg of cotton. Domestic consumers consumer all of the T-shirts produced.

The production summary table is as follows:

	Production	Prices	Labour Income	Capital Income
Cotton Farmer	150 kg	€3/kg	€400	€50
T-shirt producer	100 t-shirts	€20 each	€900	€650

- 1 Calculate the nominal GDP for this economy using the demand, income and value-added approaches.
- 2 It is assumed that the prices of cotton and T-shirts have doubled but that the quantities produced have remained the same: what is the new value of the nominal GDP? How is real GDP affected?

PS1 Exercise 1. Solutions

The production summary table is as follows:

	Production	Prices	Labour Income	Capital Income
Cotton Farmer	150 kg	€3/kg	€400	€50
T-shirt producer	100 t-shirts	€20 each	€900	€650

- ① Calculate the nominal GDP for this economy using the demand, income and value-added approaches.

► **Demand/expenditure approach:** $Y = C + I + G + NX$

- here: only consumption exists $\rightarrow C$ of final goods = consumption of t-shirts
- $GDP = 100 \text{ t-shirts bought at } €20 \text{ each} = 2000$

► **Income approach:** $Y = \text{capital income} + \text{labour income}$

- here, the labour and capital revenues given for each production
- $GDP = \underbrace{400 + 50}_{\text{Linc + Kinc for cotton}} + \underbrace{900 + 650}_{\text{Linc + Kinc for t-shirts}} = 450 + 1550 = 2000$

► **Value-Added Approach:** $Y = \sum_{\text{for each pdcer}} \left(\begin{smallmatrix} \text{all VA in that producer's} \\ \text{final gd relative to interm. inputs} \end{smallmatrix} \right)$

- here, go producer by producer and identify the input and final good
- cotton farmer: no input but produced €450 worth of cotton $\rightarrow VA_{\text{cotton}} = 450$
- t-shirt producer: input is cotton ($100 \text{ t-shirts} \times 1.5\text{kg of cotton} \times €3 = €450$) but produced €2000 worth of t-shirts $\rightarrow$
 $VA_{\text{t-shirts}} = 2000 - 450 = 1550$
- $GDP = \text{total VA} = VA_{\text{cotton}} + VA_{\text{t-shirt}} = 450 + 1550 = 2000$

PS1 Exercise 1. Solutions

The production summary table is as follows:

	Production	Prices	Labour Income	Capital Income
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- ② It is assumed that the prices of cotton and T-shirts have doubled but that the quantities produced have remained the same: what is the new value of the nominal GDP? How is real GDP affected?

- ▶ nominal GDP is measured in euros: quantity $\times$ price
- ▶ real GDP is measured independent of prices: pure quantities
- ▶ so if the price of all goods double but quantities remain unchanged, the nominal GDP doubles but the real GDP stays unchanged
- ▶ the new value of nominal GDP is €4000
- ▶ the real GDP remains at 20 shirts

PS1 Exercise 1. Three measures of GDP (cont'd)

Using the prices from Question 1, we now assume that this economy opens up to international trade. The T-shirt producer buys some of his cotton abroad and increases his production to 120 T-shirts. She exports 20 T-shirts and the remaining 100 are consumed by domestic consumers.

The new production summary table is as follows:

	Production	Prices	Labour Income	Capital Income
Domestic Cotton Farmer	100 kg	€3/kg	€275	€25
Foreign Cotton Farmer	80 kg	€3/kg	€190	€50
T-shirt producer	120 t-shirts	€20 each	€1000	€860

- ⑧ Calculate the new nominal GDP for this economy using the demand, income and value-added approaches.

PS1 Exercise 1. Solutions

The new production summary table is as follows:

	Production	Prices	Labour Income	Capital Income
Domestic Cotton Farmer	100 kg	€3/kg	€275	€25
Foreign Cotton Farmer	80 kg	€3/kg	€190	€50
T-shirt producer	$100d + 20f$ t-shirts	€20 each	€1000	€860

- ③ Calculate the new nominal GDP for this economy using the demand, income and value-added approaches.

► **Demand/expenditure approach:** $Y = C + I + G + NX$

- here: we have domestic consumption of T-shirts, import of foreign cotton and export of T-shirts
- domestic consumption: $C = 100$ t-shirts bought at €20 each = 2000
- imports: $M = 80$ kg of cotton bought at €3/kg = 240
- exports: $X = 20$ t-shirts exported and sold at €20 each = 400
- $GDP = C + I + G + X - M = 2000 + 0 + 0 + 400 - 240 = 2160$

► **Income approach:** $Y = \text{capital income} + \text{labour income}$

- here, the labour and capital revenues given for each domestic production
- $GDP = \underbrace{275 + 25}_{\text{Linc + Kinc for cotton}} + \underbrace{1000 + 860}_{\text{Linc + Kinc for t-shirts}} = 300 + 1860 = 2160$

PS1 Exercise 1. Solutions

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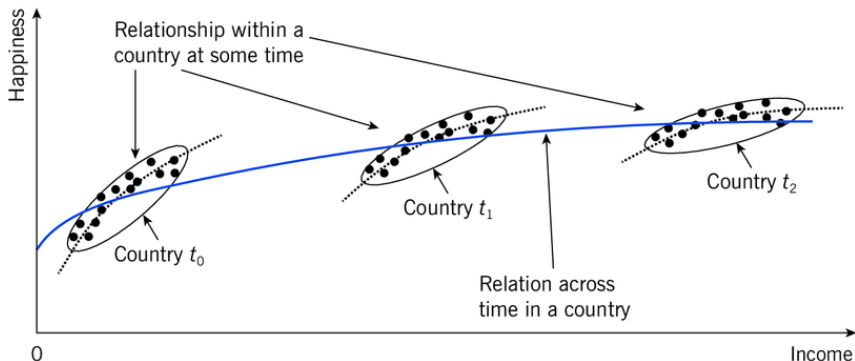
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► Value-Added Approach:

$$Y = \sum_{\text{for each dom. pdc}} \left(\begin{array}{c} \text{all VA in that producer's} \\ \text{final gd net of interm. inputs} \end{array} \right)$$

- here, go through domestic producers and identify the input and final good
- domestic cotton farmer: no input but produced € 100×3 worth of cotton → $VA_{\text{cotton}} = 300$
- t-shirt producer: input is cotton ($120 \text{ t-shirts} \times 1.5\text{kg of cotton} \times €3 = €540$) but produced € $120 \times 20 = €2400$ worth of t-shirts → $VA_{\text{t-shirts}} = 2400 - 540 = 1860$
- $GDP = \text{total VA} = VA_{\text{cotton}} + VA_{\text{t-shirt}} = 300 + 1860 = 2160$

The Easterlin Paradox

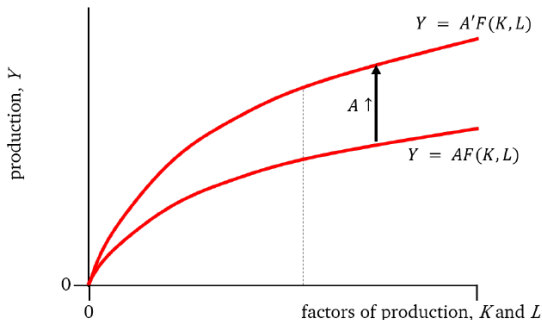


Note: In all three periods (t_0 , t_1 , t_2), higher individual income is associated with higher individual happiness (the dotted lines), but aggregate happiness does not rise along with aggregate income from t_0 to t_2 , (the solid line).

Source: Clark, A. E., P. Frijters, and M. Shields. "Relative income, happiness and utility: An explanation for the Easterlin paradox and other puzzles." *Journal of Economic Literature* 46:1 (2008): 95–144 [4].

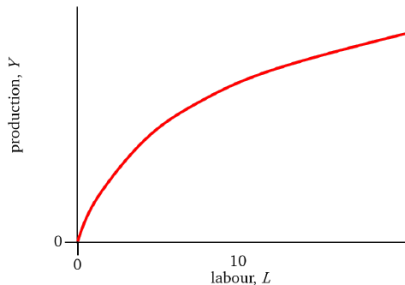
Into next week: Production functions

- ▶ GDP as production = the combination of **factors of production**: capital (K) and labour (L) + the technology that binds them all together (A)
 - ▶ Any production function needs to bring these together: $Y = F(K, L, A)$.
 - ▶ One specification is $Y = AF(K, L) \Rightarrow$ technology is an **efficiency amplifier**.
 - $\rightarrow$ for each level of K, L , an increase in A amplifies the production



Into next week: Production functions

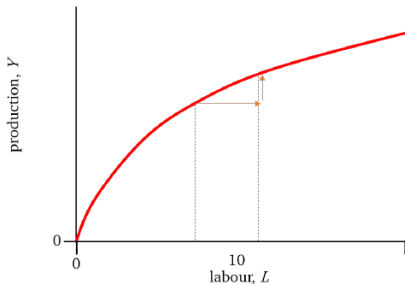
- Consider a simplified production function with just one factor of production - labour - and technology as an amplifier: $Y = AF(L)$



1. the production function is *increasing* in $L \rightarrow$ **positive marginal returns to L** : more L = more Y
2. the production function is *concave* in L :
 - $\rightarrow$ **diminishing marginal returns to L** = each additional L brings less Y than the previous
 - $\rightarrow$ **decreasing returns to scale** = more L = lower average production Y/L

Into next week: Production functions

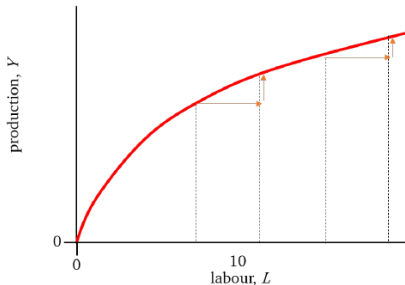
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